$(xw_i^*, w_j^*) = \overline{(xw_i, w_j)}$. Therefore, putting $P = \frac{1}{|G|}\sum_{x\in G} x$, we have

$$\begin{aligned}(t_{ij}^V, t_{i'j'}^W) &= |G|^{-1}\sum_{x\in G}(xv_i, v_j)\overline{(xw_{i'}, w_{j'})}\\ &= |G|^{-1}\sum_{x\in G}(xv_i, v_j)(xw_{i'}^*, w_{j'}^*) = (P(v_i\otimes w_{i'}^*), v_j\otimes w_{j'}^*).\end{aligned}$$

If $V \neq W$, this is zero, since P projects to the trivial representation, which does not occur in $V\otimes W^*$. If $V = W$, we need to consider $(P(v_i\otimes v_{i'}^*), v_j\otimes v_{j'}^*)$. We have a G-invariant decomposition

$$\begin{aligned}V\otimes V^* &= \mathbb{C}\oplus L,\\ \mathbb{C} &= \operatorname{span}(\sum v_k\otimes v_k^*),\\ L &= \operatorname{span}_{a:\sum_k a_{kk}=0}(\sum_{k,l} a_{kl}v_k\otimes v_l^*),\end{aligned}$$

and P projects to the first summand along the second one. The projection of $v_i\otimes v_{i'}^*$ to $\mathbb{C}\subset\mathbb{C}\oplus L$ is thus

$$\frac{\delta_{ii'}}{\dim V}\sum v_k\otimes v_k^*.$$

This shows that

$$(P(v_i\otimes v_{i'}^*), v_j\otimes v_{j'}^*) = \frac{\delta_{ii'}\delta_{jj'}}{\dim V},$$

which finishes the proof of (i) and (ii). The last statement follows immediately from the sum of squares formula. □

4.8. Character tables, examples

The characters of all the irreducible representations of a finite group can be arranged into a character table, with conjugacy classes of elements as the columns and characters as the rows. More specifically, the first row in a character table lists representatives of conjugacy classes, the second one lists the numbers of elements in the conjugacy classes, and the other rows list the values of the characters on the conjugacy classes. Due to Theorems 4.5.1 and 4.5.4, the rows and columns of a character table are orthonormal with respect to the appropriate inner products.

Note that in any character table, the row corresponding to the trivial representation consists of ones, and the column corresponding